PROBLEM 08

Relative Motion

Kinematics, Two Trains on Tracks

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Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.

PHASE 3

Step-by-Step Mathematical Deduction

Case 1: Parallel Tracks

- Train A on line $y = 0$: $\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0$.
- Train B on line $y = a$: $\mathbf{r}_B(t) = (x_{B0} + v_B t)\mathbf{i}_0 + a\mathbf{j}_0$.
- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Straight track of A along the x-axis: $\mathbf{r}_A = v_A t\,\mathbf{i}_0$.
- Circular track center at $(0, R + a)$ so that the closest distance to A's track is $(R + a) - R = a$.
- Position of train B traveling at angular speed $\omega_B = \frac{v_B}{R}$ with angle $\theta_B(t) = \omega_B t$:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\mathbf{i}_0 + [R + a - R\cos(\omega_B t)]\mathbf{j}_0$$

- Velocity of train B:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B \cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B \sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity:**

$$v_{\text{rel}} = \sqrt{\left[v_B \cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2 \sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2v_A v_B \cos\left(\frac{v_B t}{R}\right)}$$

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \text{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\text{m/s})\,\text{s}}{\text{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2v_A v_B \cos(\cdot)] = \text{m}^2/\text{s}^2$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector.
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\text{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\text{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\text{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).
- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2v_A v_B \cos(v_B t/R)$ with zero residual. Both cases are written in the single inertial basis $\mathcal{B}_0$, so no change-of-basis matrix is needed.

Source: Problems.pdf (Page 8)